

Classical column capital detection

Executive summary

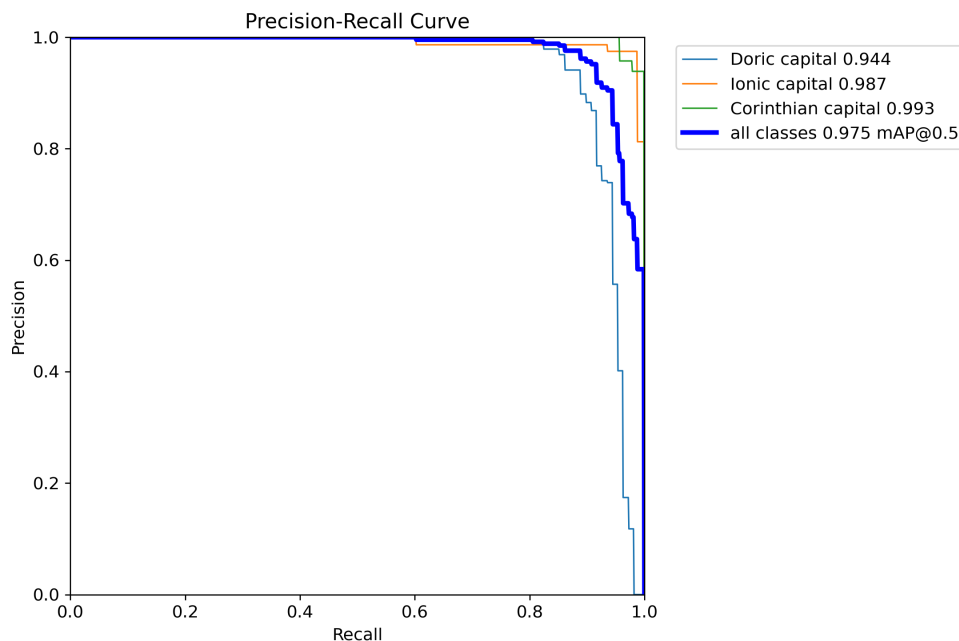
This educational AECO prototype uses a YOLOv8n object detector to locate Doric, Ionic and Corinthian column capitals in photos. It supports initial visual inventory with a person checking the results; it does not infer a whole building style.

Dataset and method

The frozen 80/20 derived split contains 862 training and 216 validation images from Roboflow Universe Architecture Segmentation version 1. Source labels were reduced to three classes and exact duplicate files removed. YOLOv8n was trained for 30 epochs at 640 pixels, batch 16, seed 42, using Ultralytics 8.4.163 and a Tesla T4. Original training lasted approximately 8.6 minutes.

Validation results

Across 232 labelled validation objects: precision 0.960; recall 0.939; mAP50 0.975; mAP50-95 0.694. Doric recall was 0.859, below Ionic 0.987 and Corinthian 0.971. These metrics describe this split and may be optimistic if related scenes appear in both subsets.



External check, limitations and next steps

What happened on five new photos

At confidence 0.25, one Ionic and one Corinthian capital were detected correctly. Two weathered or detached Doric specimens and one close-up Ionic capital produced no detection. Lowering confidence to 0.10 did not recover them. Five selected examples do not establish an overall generalization rate.

Observed errors

Validation examples show a decorative scroll misidentified as Ionic, a balcony detail boxed as Doric, and an extra duplicate Doric box around one real capital. Possible causes are lookalike ornaments and ambiguous framing; these explanations have not been tested experimentally.

Prioritized improvements

First, annotate licensed weathered and detached Doric examples. Second, add close-up Ionic images and hard negatives including scroll ornaments and balcony supports. Third, split evaluation by site, audit scene near-duplicates, retrain and compare class-wise recall plus external cases.

Governance and reproducibility

Exclude private client photos and identifiable people without permission. Check rights on every published image. Do not use this model for heritage certification, style attribution of an entire building or structural decisions. A keyless Colab notebook downloads the frozen dataset and weights from the public GitHub Release, checks the dataset SHA256 and reruns evaluation. Full 30-epoch retraining is optional.

Source dataset: universe.roboflow.com/architecture-dataset/architecture-segmentation/dataset/1

Model/release: github.com/Silvana-99/classical-capital-detection-yolo/releases/tag/v1.0

External photo authors, links and licenses: [docs/image_credits.md](#) in the repository.